Systems programming

Lec 1: Introduction

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Course Description

- This course is an introductory course on computer systems. It introduces computer systems from a programmer's perspective, rather than a system implementer's perspective, which prepares students for more advanced topics that discuss the internals of a computer system (e.g., operating systems or computer architecture).
- As a result, the focus of the course is teaching programmable interfaces of a computer system as well as how to use them correctly and effectively when writing a program.

Course objectives

- This course is concerned with systems programming.
- You will learn:
  - More about the properties of computer systems
  - How architecture affects programs
  - How to effectively write efficient and correct programs
  - The C programming language and POSIX API
- Our goals are to:
  - improve your intuitions about how computers work
  - Practice solving real problems with programming

Course topics

- The topics mainly include:
  - Hardware/software interfaces (e.g., data representation in memory)
  - OS/application interfaces (e.g., syscalls).
- In discussing these topics, the course gives an overview of a complete computer system, the hardware, operating system, compiler, and network, in order to guide students through various components that modern programs rely on to accomplish their intended purposes.

Course references

- There are two required texts for this course. Students are expected to have immediate access to both of these texts throughout the semester.
  - Computer Systems: A Programmer's Perspective (Third Edition), by Bryant and O'Hallaron.
  - The C Programming Language (Second Edition), by Kernighan and Ritchie.

* Lecture slides will be provided electronically throughout the course of the semester.

Prerequisites

- Students are expected to have a solid grasp of simple data structures (such as lists and arrays), some programming experience and good background in operating systems concepts.

Contents

- The topics to be covered in this course include:
  - The C programming language
  - Compiler and linker invocation and their roles in program development
  - Version control, build, and other systems
  - Representation of in-memory data structures
  - Memory management
  - The kernel/userspace boundary and the system call interface
  - Threads
  - Synchronization primitives and concurrency

Course grading

- Assignments and quizzes 20%.
- Practical part (On lab) 30%.
- Final exam 50%.

Why C?

- There are dozens of programming languages. Why C?
  - C is “high level” — but not very.
  - C provides functions, structured programming, complex data types, and many other powerful abstractions
  - ...yet it also exposes many architectural details
  - Most operating system kernels are written in C.
  - Many runtimes and virtual machines are written in C.
  - C influences many other languages.

The Processor and Memory

- The C language exposes a particular machine model.
- Data is stored in memory at accessible addresses.
- The CPU manipulates data stored in memory.
- Program code is executed as a series of instructions:
  - Also stored in memory
  - Though possibly not accessible (as data)

A Dedicated Computer

- Most modern, multitasking OSes (including Unix) provide a particular model.
- That model is that each process has its own dedicated machine.
- Each process appears to have:
  - A dedicated CPU
  - Private, dedicated memory
  - Private input and output facilities
- That isn't strictly true, but it is approximated by the OS.
- The OS provides mechanisms to share resources in this model.

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Programs as Instructions

- C programs are translated into machine instructions.
- The computer executes these instructions in order.
- Instructions are things like:
  - Add two numbers together
  - Compare a number to zero
  - Store a number to a location in memory

Developing Hello World

- “Hello World” is a classic first program when learning a language.
- We will develop a Hello World together.

Summary

- C is a high level language used in systems programming.
- Architectural details are important in C.
- The C/POSIX model is:
  - A dedicated machine for each program
  - Sequential execution of program instructions
  - Data is stored in accessible, addressed memory

Next Time …

- More about types
- Variable declaration and usage
- C Strings
- Looping

References

- 2023 Ethan Blanton & Carl Alphonce / CSE 220: Systems Programming course